

Google GenAI setup — models, parameters

Ten components call Gemini via the Google GenAI SDK: three **agents** (multi-turn tool-use loops) and seven **one-shot callers** (single API call). Agents use the Pro tier for superior visual reasoning; one-shot callers use the Flash tier for speed, except for the translator which uses a dedicated lite model.

Authentication

The system uses **API Key authentication**. Ensure your key is exported in your environment:

```
export GEMINI_API_KEY=your-api-key-here
```

Callers

Caller	Pipeline step	Default model
macro-images agent	Pass 1: Macro-Object Discovery	gemini-3.1-pro-preview
micro-images agent	Pass 2: Micro-Detail Subdivision	gemini-3.1-pro-preview
re-crop agent	programmatic region refinement (on quality failures)	gemini-3.1-pro-preview
verifier	programmatic quality checking (zero-turn)	gemini-3.5-flash (pinned)
layer-proposer	document layer approval	gemini-3.5-flash
captioner	caption generation	gemini-3.5-flash
translator	index building (per non-English text block)	gemini-3.1-flash-lite

Caller	Pipeline step	Default model
layer-assigner	asset layer assignment	gemini-3.5-flash
query-expander	query expansion (optional)	gemini-3.5-flash
answer-synthesizer	answer synthesis	gemini-3.5-flash

The verifier is pinned to Flash by design — a less-capable model judges the more-capable agents (independence by asymmetry). The other one-shot callers use Flash because they don't iterate. Technical translation is handled by a dedicated lite model to optimize throughput and cost during large-scale index builds.

Agent parameters

macro-images & micro-images agents

- `MAX_TURNS = 100` per page session (long-context visual reasoning)
- `MAX_OUTPUT_TOKENS = 8192`
- `TEMPERATURE = 0.2` (Broad nucleus for wide spatial visual exploration)
- `TOP_P = 0.85`
- CLI flags for macro-images & micro-images: `--model {pro,flash}` , `--workers N`

re-crop agent

- `MAX_TURNS = 50` per region session (one session per failed region, capped at 50 turns to allow gradual coordination)
- `MAX_OUTPUT_TOKENS = 8192`
- `TEMPERATURE = 0.1` (Surgically clamped nucleus for gradual, progressive coordinate modulation)
- `TOP_P = 0.15`

- Runs programmatically and internally inside the propose commands; permanently pinned to `gemini-3.1-pro-preview` for superior coordinate nudging, regardless of the parent agent's model tier.

verifier

- `MAX_ROUNDS = 3` verify→re-crop cycles before remaining failures are permanently rejected (enables highly granular, gradual improvements)
- `MAX_OUTPUT_TOKENS = 8192`
- Loads the entire suite of cropping rules from the corresponding Pass (Macro or Micro) in `docs/rules.md` to guarantee 100% evaluation symmetry.
- Runs programmatically and internally inside the propose commands (MIME-type synced to `image/png` to prevent HTTP body leaks).
- Runs on `gemini-3.5-flash` (pinned by design to evaluate Pro-generated drafts)

One-shot caller limits

- `CAPTION_MAX_OUTPUT_TOKENS = 2048`
- `SYNTHESIZE_MAX_OUTPUT_TOKENS = 8192`
- `QUERY_EXPAND_MAX_OUTPUT_TOKENS = 4096`
- `PROPOSE_LAYER_MAX_OUTPUT_TOKENS = 4096`
- `ASSIGN_LAYER_MAX_OUTPUT_TOKENS = 4096`
- `TRANSLATE_MAX_OUTPUT_TOKENS = 8192`

Use cases

Use case	Setting
Default production	<code>gemini-3.1-pro-preview</code> for the three agents (macro-images, micro-images, and re-crop), <code>gemini-3.5-flash</code> for the six general one-shot callers; <code>gemini-3.1-flash-lite</code> for translation
Cheaper A/B on agents	<code>--model flash</code> during region proposal (<code>macro-images</code> or <code>micro-images</code>) — agents drop to Flash; one-shot callers unchanged
Forced Layer Re-evaluation	<code>rag asset-layers --force</code> — resets layer verification flags and forces re-evaluation of all assets from scratch
Batch Query Processing	<code>rag query --batch</code> — batch processes queries from <code>library/queries.rtf</code> and writes answers directly to disk

Caching

The propose-agent and re-crop agent wrap their system prompt with an ephemeral prompt-cache marker (1-hour TTL). Inside the 1-hour window the rules text is billed at ~10% of the input rate; the one-time cache-write penalty (~2× input rate) is recouped after ~9 reads. The verifier and captioner do not use caching — their system prompts are below the minimum cacheable prefix for Flash models, so a marker would silently no-op.